

EYES AND VISION

EYESIGHT PROVIDES THE BRAIN WITH MORE INPUT THAN ALL OTHER SENSES COMBINED – MORE THAN HALF THE INFORMATION IN THE CONSCIOUS MIND IS ESTIMATED TO ENTER THROUGH THE EYES.

THE SEQUENCE OF VISION

Light rays enter the eye through the clear, domed front of the eyeball, the cornea, where they are partly bent (refracted). They then pass through the transparent lens, which changes shape to focus them (see Accommodation, right) as an upside-down image onto the retina. The retina contains many millions of light-sensitive cells called rods and cones, which convert light energy into nerve signals. Rods are scattered through the retina and detect low levels of light. Cones are concentrated in the fovea and distinguish colours and fine details. The signals of the image are sent along the optic nerves of each eye to the visual cortex in the brain.

Optic nerve
Conveys nerve signals to brain

Lateral rectus
Small muscle that swivels the eye to look out to the side

INSIDE THE EYE

An average eyeball is 25mm (1in) in diameter, and has three main outer layers: the sclera, choroid, and retina. Near the front, the sclera can be seen as the white of the eye, and at the front it becomes the clear cornea. The main bulk of the eye, between the lens and the retina, is filled with a clear, jelly-like fluid known as vitreous humour. This maintains the eyeball's spherical shape.

Sclera
Tough, white, protective outer sheath of eyeball

Choroid
Blood-rich layer that supplies retina and sclera

Retina
Thin layer of light-sensitive rod and cone cells

Fovea
Region of retina with dense concentration of cone cells, enabling precise vision

Optic disc
Point at which nerve fibres leave the eye; contains no light-sensitive cells